Lesson 22 material: Conducting Surveys

7.4: Conducting Surveys

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Learning Objectives

1. Explain the difference between probability and non-probability sampling, and describe the major types of probability sampling.
2. Define sampling bias in general and non-response bias in particular. List some techniques that can be used to increase the response rate and reduce non-response bias.
3. List the four major ways to conduct a survey along with some pros and cons of each.

In this section, we consider how to go about conducting a survey. We first consider the issue of sampling, followed by some different methods of actually collecting survey data.

Sampling

Essentially all psychological research involves sampling—selecting a sample to study from the population of interest. Sampling falls into two broad categories. The first category, **Probability sampling** , occurs when the researcher can specify the probability that each member of the population will be selected for the sample. The second is **Non-probability sampling** , which occurs when the researcher cannot specify these probabilities. Most psychological research involves non-probability sampling. For example, **Convenience sampling**—studying individuals who happen to be nearby and willing to participate—is a very common form of non-probability sampling used in psychological research. Other forms of non-probability sampling include **snowball sampling** (in which existing research participants help recruit additional participants for the study), **quota sampling** (in which subgroups in the sample are recruited to be proportional to those subgroups in the population), and **self-selection sampling** (in which individuals choose to take part in the research on their own accord, without being approached by the researcher directly).

Survey researchers, however, are much more likely to use some form of probability sampling. This tendency is because the goal of most survey research is to make accurate estimates about what is true in a particular population, and these estimates are most accurate when based on a probability sample. For example, it is important for survey researchers to base their estimates of election outcomes—which are often decided by only a few percentage points—on probability samples of likely registered voters.

Compared with non-probability sampling, probability sampling requires a very clear specification of the population, which of course depends on the research questions to be answered. The population might be all registered voters in Washington State , all American consumers who have purchased a car in the past year, women in the Seattle over 40 years old who have received a mammogram in the past decade, or all the alumni of a particular university. Once the population has been specified, probability sampling requires a **sampling frame** . This sampling frame is essentially a list of all the members of the population from which to select the respondents. Sampling frames can come from a variety of sources, including telephone directories, lists of registered voters, and hospital or insurance records. In some cases, a map can serve as a sampling frame, allowing for the selection of cities, streets, or households.

There are a variety of different probability sampling methods. **Simple random sampling** is done in such a way that each individual in the population has an equal probability of being selected for the sample. This type of sampling could involve putting the names of all individuals in the sampling frame into a hat, mixing them up, and then drawing out the number needed for the sample. Given that most sampling frames take the form of computer files, random sampling is more likely to involve computerized sorting or selection of respondents. A common approach in telephone surveys is random-digit dialing, in which a computer randomly generates phone numbers from among the possible phone numbers within a given geographic area.

A common alternative to simple random sampling is **stratified random sampling** , in which the population is divided into different subgroups or "strata" (usually based on demographic characteristics) and then a random sample is taken from each "stratum." **Proportionate stratified random sampling** can be used to select a sample in which the proportion of respondents in each of various subgroups matches the proportion in the population. For example, because about 12.6 % of the American population is African American , stratified random sampling can be used to ensure that a survey of 1,000 American adults includes about 126 African-American respondents. **Disproportionate stratified random sampling** can also be used to sample extra respondents from particularly small subgroups—allowing valid conclusions to be drawn about those subgroups. For example, because Asian Americans make up a relatively small percentage of the American population (about 5.6 %), a simple random sample of 1,000 American adults might include too few Asian Americans to draw any conclusions about them as distinct from any other subgroup. If representation is important to the research question, however, then disproportionate stratified random sampling could be used to ensure that enough Asian-American respondents are included in the sample to draw valid conclusions about Asian American s a whole.

Yet another type of probability sampling is **cluster sampling**, in which larger clusters of individuals are randomly sampled and then individuals within each cluster are randomly sampled. This is the only probability sampling method that does not require a sampling frame. For example, to select a sample of small-town residents in Washington , a researcher might randomly select several small towns and then randomly select several individuals within each town. Cluster sampling is especially useful for surveys that involve face-to-face interviewing because it minimizes the amount of traveling that the interviewers must do. For example, instead of traveling to 200 small towns to interview 200 residents, a research team could travel to 10 small towns and interview 20 residents of each. The National Comorbidity Survey was done using a form of cluster sampling.

How large does a survey sample need to be? In general, this estimate depends on two factors. One is the level of confidence in the result that the researcher wants. The larger the sample, the closer any statistic based on that sample will tend to be to the corresponding value in the population. The other factor is a practical constraint in the form of the budget of the study. Larger samples provide greater confidence, but they take more time, effort, and money to obtain. Taking these two factors into account, most survey research uses sample sizes that range from about 100 to about 1,000. Conducting a power analysis prior to launching the survey helps to guide the researcher in making this trade-off.

Sample Size and Population Size

Why is a sample of about 1,000 considered to be adequate for most survey research—even when the population is much larger than that? Consider, for example, that a sample of only 1,000 American adults is generally considered a good sample of the roughly 252 million adults in the American population—even though it includes only about 0.000004% of the population! The answer is a bit surprising.

One part of the answer is that a statistic based on a larger sample will tend to be closer to the population value and that this can be characterized mathematically. Imagine, for example, that in a sample of registered voters, exactly 50% say they intend to vote for the incumbent. If there are 100 voters in this sample, then there is a 95% chance that the true percentage in the population is between 40 and 60. But if there are 1,000 voters in the sample, then there is a 95% chance that the true percentage in the population is between 47 and 53. Although this "95% confidence interval" continues to shrink as the sample size increases, it does so at a slower rate. For example, if there are 2,000 voters in the sample, then this reduction only reduces the 95% confidence interval to 48 to 52. In many situations, the small increase in confidence beyond a sample size of 1,000 is not considered to be worth the additional time, effort, and money.

Another part of the answer—and perhaps the more surprising part—is that confidence intervals depend only on the size of the sample and not on the size of the population. So a sample of 1,000 would produce a 95% confidence interval of 47 to 53 regardless of whether the population size was a hundred thousand, a million, or a hundred million.

Sampling Bias

Probability sampling was developed in large part to address the issue of sampling bias. **Sampling bias** occurs when a sample is selected in such a way that it is not representative of the entire population and therefore produces inaccurate results. This bias was the reason that the *Literary Digest* straw poll was so far off in its prediction of the 1936 presidential election. The mailing lists used came largely from telephone directories and lists of registered automobile owners, which over-represented wealthier people, who were more likely to vote for Landon. Gallup was successful because he knew about this bias and found ways to sample less wealthy people as well.

There is one form of sampling bias that even careful random sampling is subject to. It is almost never the case that everyone selected for the sample actually responds to the survey. Some may have died or moved away, and others may decline to participate because they are too busy, are not interested in the survey topic, or do not participate in surveys on principle. If these survey non-responders differ from survey responders in systematic ways, then this difference can produce **non-response bias** . For example, in a mail survey on alcohol consumption, researcher Vivienne Lahaut and colleagues found that only about half the sample responded after the initial contact and two follow-up reminders (Lahaut, Jansen, van de Mheen, & Garretsen, 2002) [1]. The danger here is that the half who responded might have different patterns of alcohol consumption than the half who did not, which could lead to inaccurate conclusions on the part of the researchers. So to test for non-response bias, the researchers later made unannounced visits to the homes of a subset of the non-responders—coming back up to five times if they did not find them at home. They found that the original non-responders included an especially high proportion of abstainers (nondrinkers), which meant that their estimates of alcohol consumption based only on the original responders were too high.

Although there are methods for statistically correcting for non-response bias, they are based on assumptions about the non-responders—for example, that they are more similar to late responders than to early responders—which may not be correct. For this reason, the best approach to minimizing non-response bias is to minimize the number of non-responders—that is, to maximize the response rate. There is a large research literature on the factors that affect survey response rates (Groves et al., 2004) [2]. In general, in-person interviews have the highest response rates, followed by telephone surveys, and then mail and Internet surveys. Among the other factors that increase response rates are sending potential respondents a short pre-notification message informing them that they will be asked to participate in a survey in the near future and sending simple follow-up reminders to non-responders after a few weeks. The perceived length and complexity of the survey can also make a difference, which is why it is important to keep survey questionnaires as short, simple, and on topic as possible. Finally, offering an incentive—especially cash—is a reliable way to increase response rates. However, ethically, there are limits to offering incentives that may be so large as to be considered coercive.

Conducting the Survey

The four main ways to conduct surveys are through in-person interviews, by telephone, through the mail, and over the internet. As with other aspects of survey design, the choice depends on both the researcher's goals and the budget. In-person interviews have the highest response rates and provide the closest personal contact with respondents. Personal contact can be important, for example, when the interviewer must see and make judgments about respondents, as is the case with some mental health interviews. But in-person interviewing is by far the most costly approach. Telephone surveys have lower response rates and still provide some personal contact with respondents. They can also be costly but are generally less so than in-person interviews. Traditionally, telephone directories have provided fairly comprehensive sampling frames. However, this trend is less true today as more people choose to only have cell phones and do not install land lines that would be included in telephone directories. Mail surveys are less costly still but generally have even lower response rates—making them most susceptible to non-response bias.

Not surprisingly, internet surveys are becoming more common. They are increasingly easy to construct and use (see "Online Survey Creation"). Although initial contact can be made by mail with a link provided to the survey, this approach does not necessarily produce higher response rates than an ordinary mail survey. A better approach is to make initial contact by email with a link directly to the survey. This approach can work well when the population consists of the members of an organization who have known email addresses and regularly use them (e.g., a university community). For other populations, it can be difficult or impossible to find a comprehensive list of email addresses to serve as a sampling frame. Alternatively, a request to participate in the survey with a link to it can be posted on websites known to be visited by members of the population. But again it is very difficult to get anything

approaching a random sample this way because the members of the population who visit the websites are likely to be different from the population as a whole. However, internet survey methods are in rapid development. Because of their low cost, and because more people are online than ever before, internet surveys are likely to become the dominant approach to survey data collection in the near future.

Finally, it is important to note that some of the concerns that people have about collecting data online (e.g., that internet-based findings differ from those obtained with other methods) have been found to be myths. Figure (adapted from Gosling, Vazire, Srivastava, & John, 2004) [3] addresses three such preconceptions about data collected in web-based studies:

Figure: Some Preconceptions and Findings Pertaining to Web-based Studies

Preconception	**Finding**
Internet samples are not demographically diverse	Internet samples are more diverse than traditional samples in many domains, although they are not completely representative of the population
Internet samples are maladjusted, socially isolated, or depressed	Internet users do not differs from nonusers on markers of adjustment and depression
Internet-based findings differ from those obtained with other methods	Evidence so far suggests that internet-based findings are consistent with findings based on traditional methods (e.g., on self-esteem, personality), but more data are needed.

Online Survey Creation

There are now several online tools for creating online questionnaires. After a questionnaire is created, a link to it can then be emailed to potential respondents or embedded in a web page. The following websites are among those that offer free accounts. Although the free accounts limit the number of questionnaire items and the number of respondents, they can be useful for doing small-scale surveys and for practicing the principles of good questionnaire construction. Here are some commonly used online survey tools:

- SurveyMonkey—https://surveymonkey.com
- PsyToolkit—https://www.psytoolkit.org/ (free, noncommercial, and does many experimental paradigms)
- Qualtrics—https://www.qualtrics.com/
- PsycData—https://www.psychdata.com/

A small note of caution: the data from US survey software are held on US servers, and are subject to be seized as granted through the Patriot Act. To avoid infringing on any rights, the following is a list of online survey sites that are hosted in Canada:

- Fluid Surveys—http://fluidsurveys.com/
- Simple Survey—http://www.simplesurvey.com/
- Lime Survey—https://www.limesurvey.org

There are also survey sites hosted in other countries outside of North America.

Another new tool for survey researchers is Mechanical Turk (MTurk) created by Amazon.com https://www.mturk.com Originally created for simple usability testing, MTurk has a database of over 500,000 workers from over 190 countries[4]. You can put simple tasks (for example, different question wording to test your survey items), set parameters as your sample frame dictates and deploy your experiment at a very low cost (for example, a few cents for less than 5 minutes). MTurk has been lauded as an inexpensive way to gather high-quality data (Buhrmester, Kwang, & Gosling, 2011)[5].

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7.5: Survey Research (Summary)

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Key Takeaways

- Survey research features the use of self-report measures on carefully selected samples. It is a flexible approach that can be used to study a wide variety of basic and applied research questions.
- Survey research has its roots in applied social research, market research, and election polling. It has since become an important approach in many academic disciplines, including political science, sociology, public health, and, of course, psychology.
- Survey research involves asking respondents to self-report on their own thoughts, feelings, and behaviors.
- Most survey research is non-experimental in nature (it is used to describe variables or measure statistical relationships between variables) but surveys can also be used to measure dependent variables in true experiments.
- Responding to a survey item is itself a complex cognitive process that involves interpreting the question, retrieving information, making a tentative judgment, putting that judgment into the required response format, and editing the response.
- Survey responses are subject to numerous context effects due to question wording, item order, response options, and other factors. Researchers should be sensitive to such effects when constructing surveys and interpreting survey results.
- Survey items are either open-ended or closed-ended. Open-ended items simply ask a question and allow respondents to answer in whatever way they want. Closed-ended items ask a question and provide several response options that respondents must choose from.
- Use verbal labels instead of numerical labels although the responses can be converted to numerical data in the analyses.
- According to the BRUSO model, questionnaire items should be brief, relevant, unambiguous, specific, and objective.
- Survey research usually involves probability sampling, in which each member of the population has a known probability of being selected for the sample. Types of probability sampling include simple random sampling, stratified random sampling, and cluster sampling.
- Sampling bias occurs when a sample is selected in such a way that it is not representative of the population and therefore produces inaccurate results. The most pervasive form of sampling bias is non-response bias, which occurs when people who do not respond to the survey differ in important ways from people who do respond. The best way to minimize non-response bias is to maximize the response rate by prenotifying respondents, sending them reminders, constructing questionnaires that are short and easy to complete, and offering incentives.
- Surveys can be conducted in person, by telephone, through the mail, and on the internet. In-person interviewing has the highest response rates but is the most expensive. Mail and internet surveys are less expensive but have much lower response rates. Internet surveys are likely to become the dominant approach because of their low cost.

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Exercises

- Discussion: Think of a question that each of the following professionals might try to answer using survey research.
  - a social psychologist
  - an educational researcher
  - a market researcher who works for a supermarket chain
  - the mayor of a large city
  - the head of a university police force
- Discussion: Write a survey item and then write a short description of how someone might respond to that item based on the cognitive model of survey responding (or choose any item on the Rosenberg Self-Esteem Scale at http://www.bsos.umd.edu/socy/research/rosenberg.htm).
- Practice: Write survey items for each of the following general questions. In some cases, a series of items, rather than a single item, might be necessary.
  - How much does the respondent use Facebook?
  - How much exercise does the respondent get?
  - How likely does the respondent think it is that the incumbent will be re-elected in the next presidential election?
  - To what extent does the respondent experience “road rage”?
- Discussion: If possible, identify an appropriate sampling frame for each of the following populations. If there is no appropriate sampling frame, explain why.
  - students at a particular university
  - adults living in the state of Washington
  - households in Pullman, Washington
  - people with low self-esteem
- Practice: Use one of the online survey creation tools to create a 10-item survey questionnaire on a topic of your choice.